

Element F: Consideration of Design Viability

When designing the way the water filter would look and stand I had to make sure that it could stand up on its own without falling over. In order to do this I would have to make the base be larger than the top since it would be top heavy from the materials. Then I would have to make sure the base would not collapse from the weight that would be on it. I decided to use a one gallon bottle that would sit under the filter and cut some of the top off of the bottle to make sure the filter would fit snugly into the bottle and not tip over. By doing this the filter would not tip over and the bottle under the filter would catch the filtered water so it wouldn't drip out everywhere.

As far as the filter itself I just had to make sure everything was in the right order and make sure nothing would fall out of the bottom. I did this by using cotton and a piece of cloth at the bottom of the filter where the lid goes on and making sure the cloth was over the hole at the bottom. I then used a rubber band to put around the cloth and lid so the cloth would stay and not fall off. By doing this none of the materials would fall out of the bottom. This would ensure that as long as the rubber band was on tightly, only water would come out of the bottom.